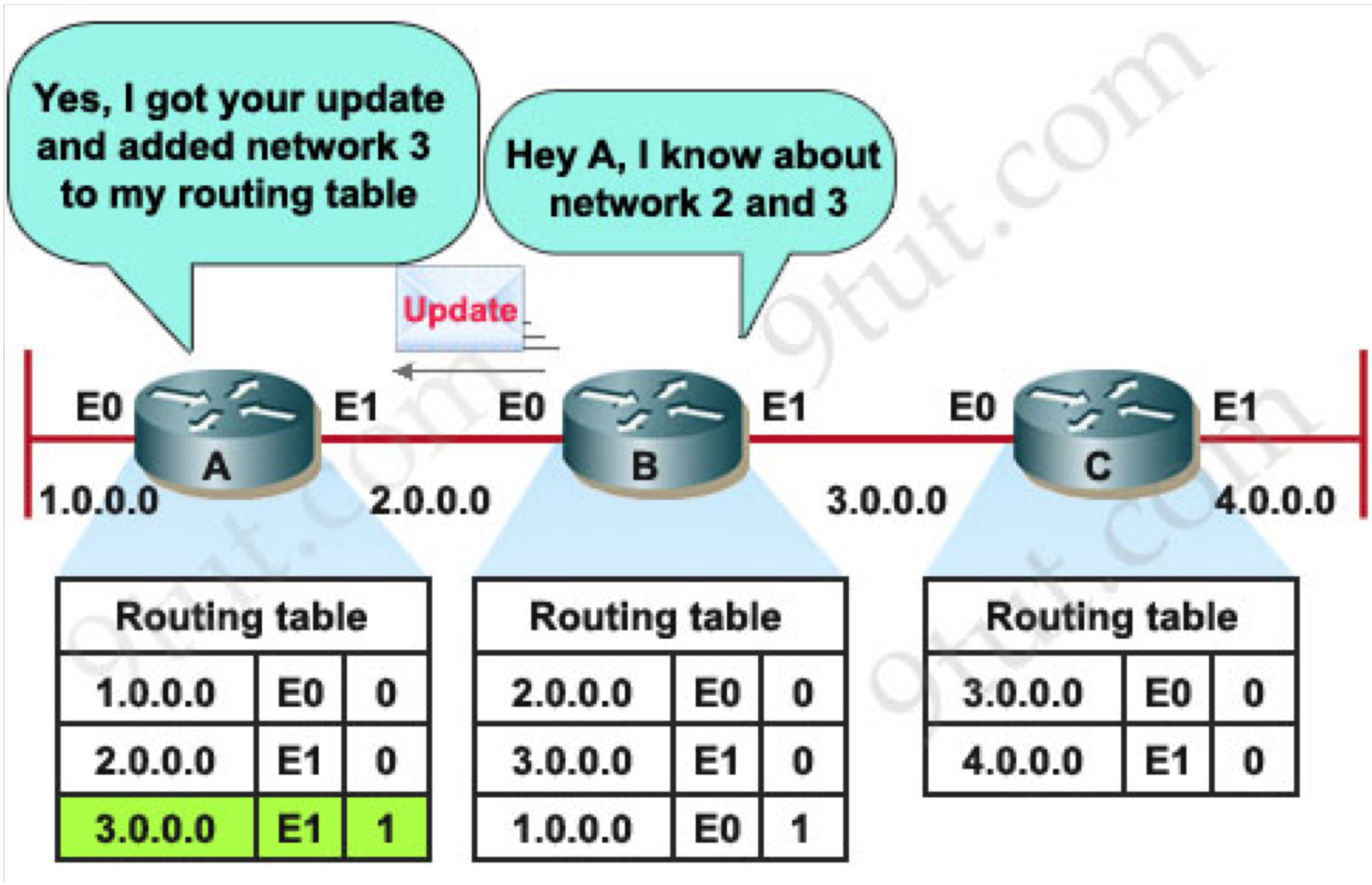
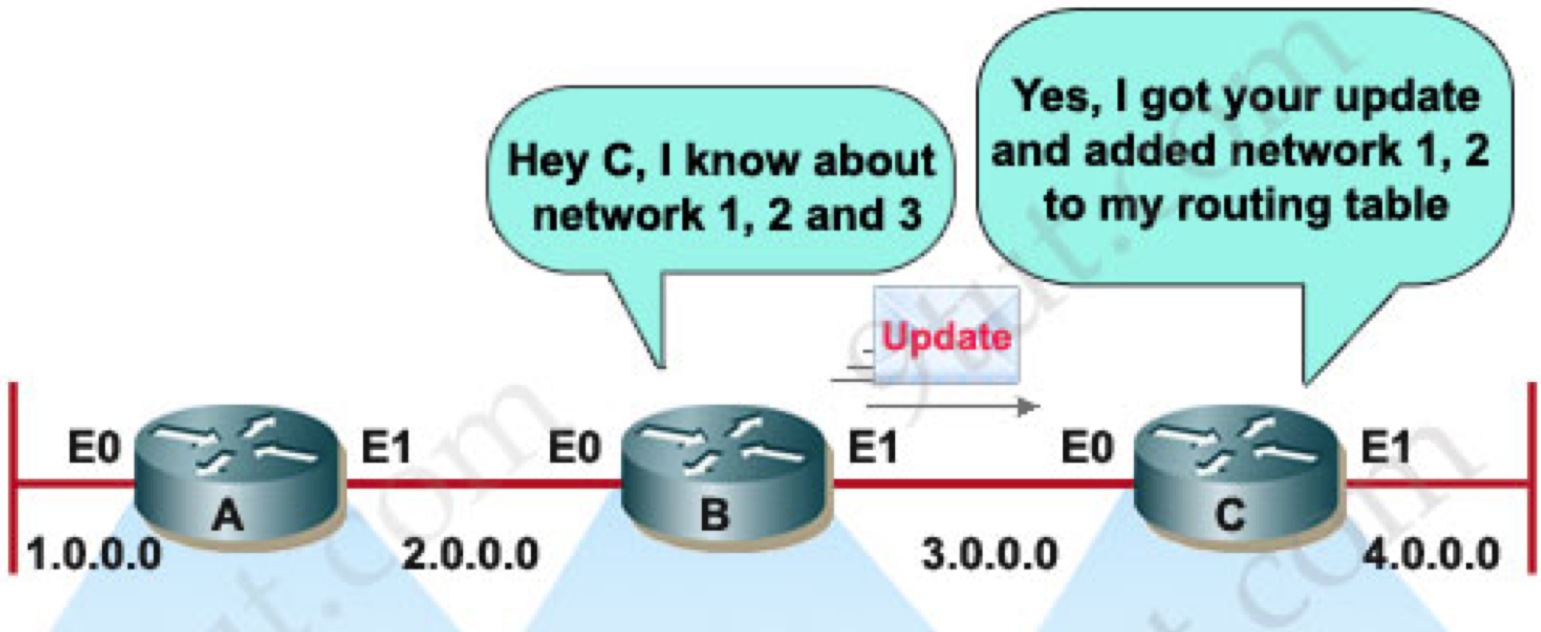
After Second Update



After Third Update

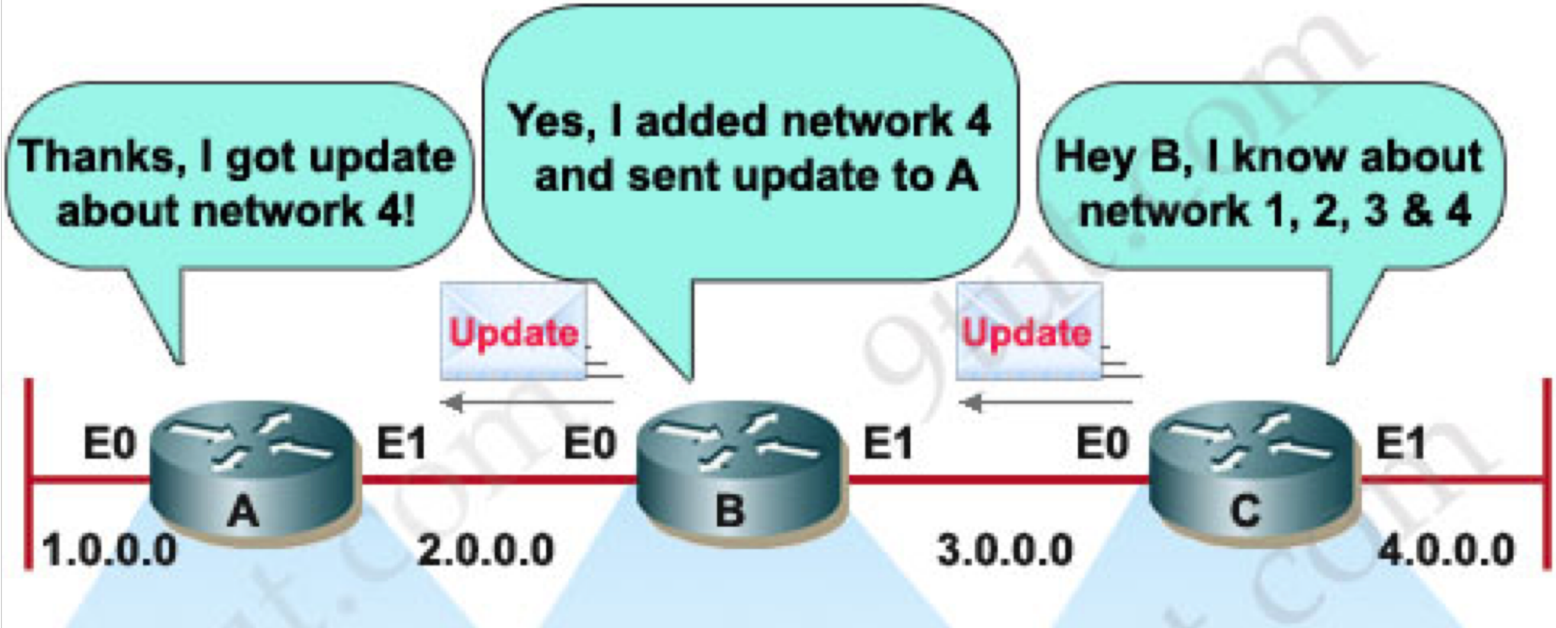


Routing table		
1.0.0.0	E0	0
2.0.0.0	E1	0
3.0.0.0	E1	1

Routing table		
2.0.0.0	E0	0
3.0.0.0	E1	0
1.0.0.0	E0	1

Routing table		
3.0.0.0	E0	0
4.0.0.0	E1	0
1.0.0.0	E0	2
2.0.0.0	E0	1

Last Update for Convergence



Routing table		
1.0.0.0	E0	0
2.0.0.0	E1	0
3.0.0.0	E1	1
4.0.0.0	E1	2

Routing table		
2.0.0.0	E0	0
3.0.0.0	E1	0
1.0.0.0	E0	1
4.0.0.0	E1	1

Routing table		
3.0.0.0	E0	0
4.0.0.0	E1	0
1.0.0.0	E0	2
2.0.0.0	E0	1

Convergence and Loops

- Distance Vector Protocols are subject to loop formations because of the myopic view of each router.
  - routers only hear from neighbors and use that to create a global connectivity map.
- when changes occur, they are broadcast but take a while to propagate and during that time cycles can form.
- one particular problem is the count to infinity problem, where updates bounce back and forth and the distance or cost creeps up in value.
- to counter that, a maximum value is set that once it is reached, the destination is considered to be unreachable and the route is removed from the routing table.

RIP Summary and demise

- low overhead – fully distributed … BUT
- slow convergence
- low overhead
- limited to 15 hops (max path cost → infinity =16)
- only uses local information from immediate neighbors for routing decisions - relies on propagation of information for global view of network – cycle formations
- No longer used – Rest in Peace (RIP)

OSPF (Open Shortest Path First)

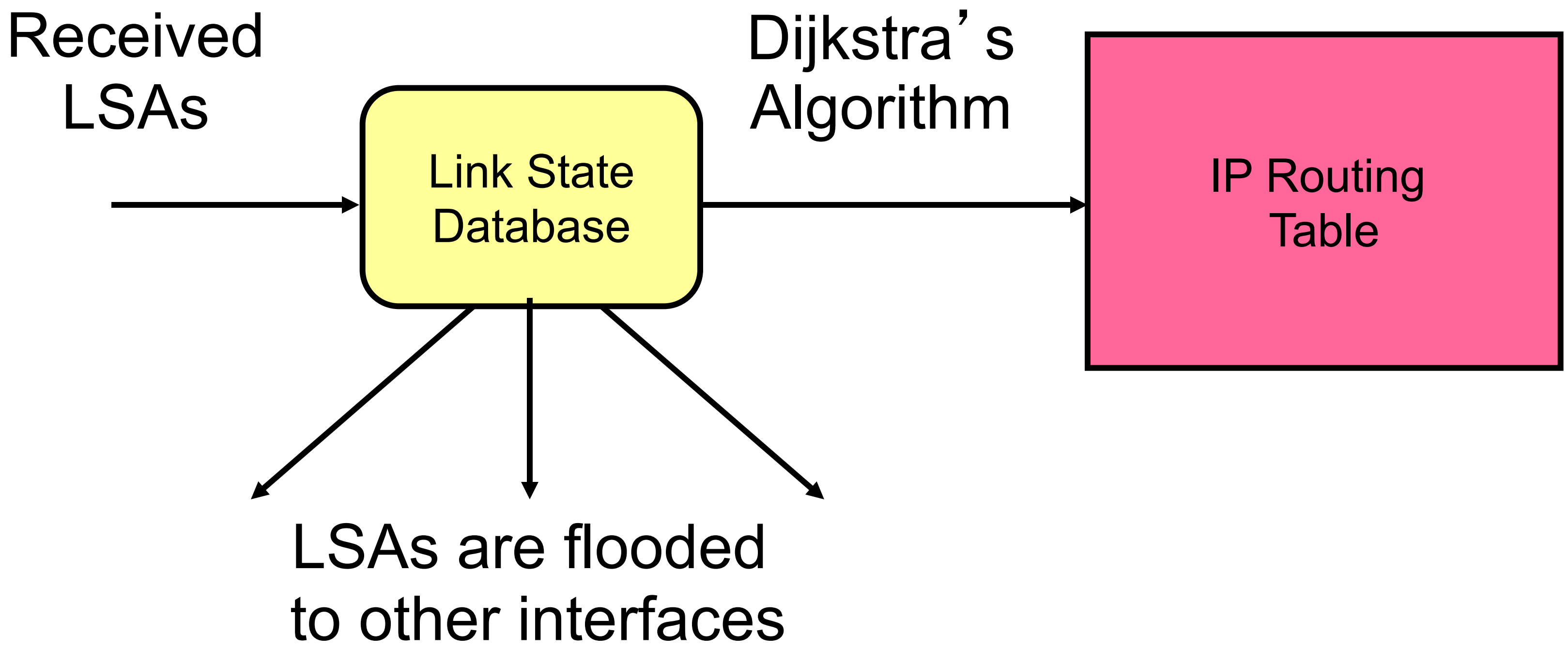
- "open": publicly available
- uses link-state algorithm
  - link state packet dissemination
  - topology map at each node
  - route computation using Dijkstra's algorithm
- router floods OSPF link-state advertisements to all other routers in *entire* AS
  - carried in OSPF messages directly over IP (rather than TCP or UDP – protocol type 89
  - link state: for each attached link

OSPF: Basic principles

- routers establish a relationship *("adjacency")* with neighbors
- each router generates *link state advertisements (LSAs)* which are distributed to all "adjacent" routers (after routers have established adjacencies).

  LSA = (link id, state of the link, cost, neighbors of the link)
- each router maintains a database (LSDB). consists of all received LSAs (*topological database* or *link state database*), which describes the network as a graph with weighted edges
- each router uses its link state database to run a shortest path algorithm (Dijikstra's algorithm) to produce the shortest path to each network

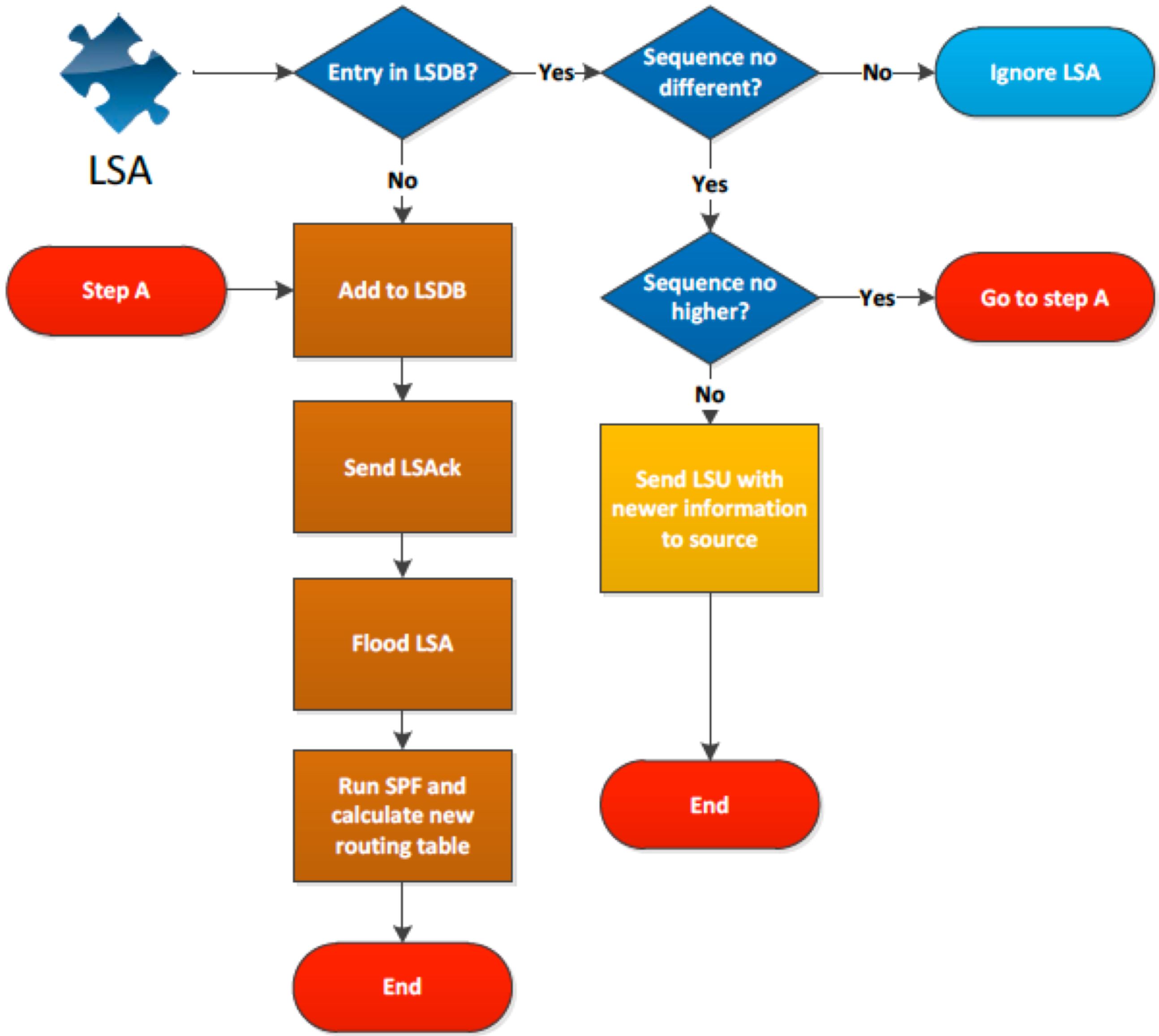
Operation of a Link State Routing protocol



LSA Updates

- link-state routing protocols generate routing updates only when a change occurs in the network topology.
- when a link changes state, the device that detected the change creates a link-state advertisement (LSA) concerning that link and sends it to all neighboring devices using a special multicast address.
- each routing device reads the LSA.
  - the LSA has a sequence number that allows the router to check to see if it has already seen that update
  - if old, it is discarded, if new, link-state database (LSDB) info updated and LSA passed along to next neighbors

Flow Chart



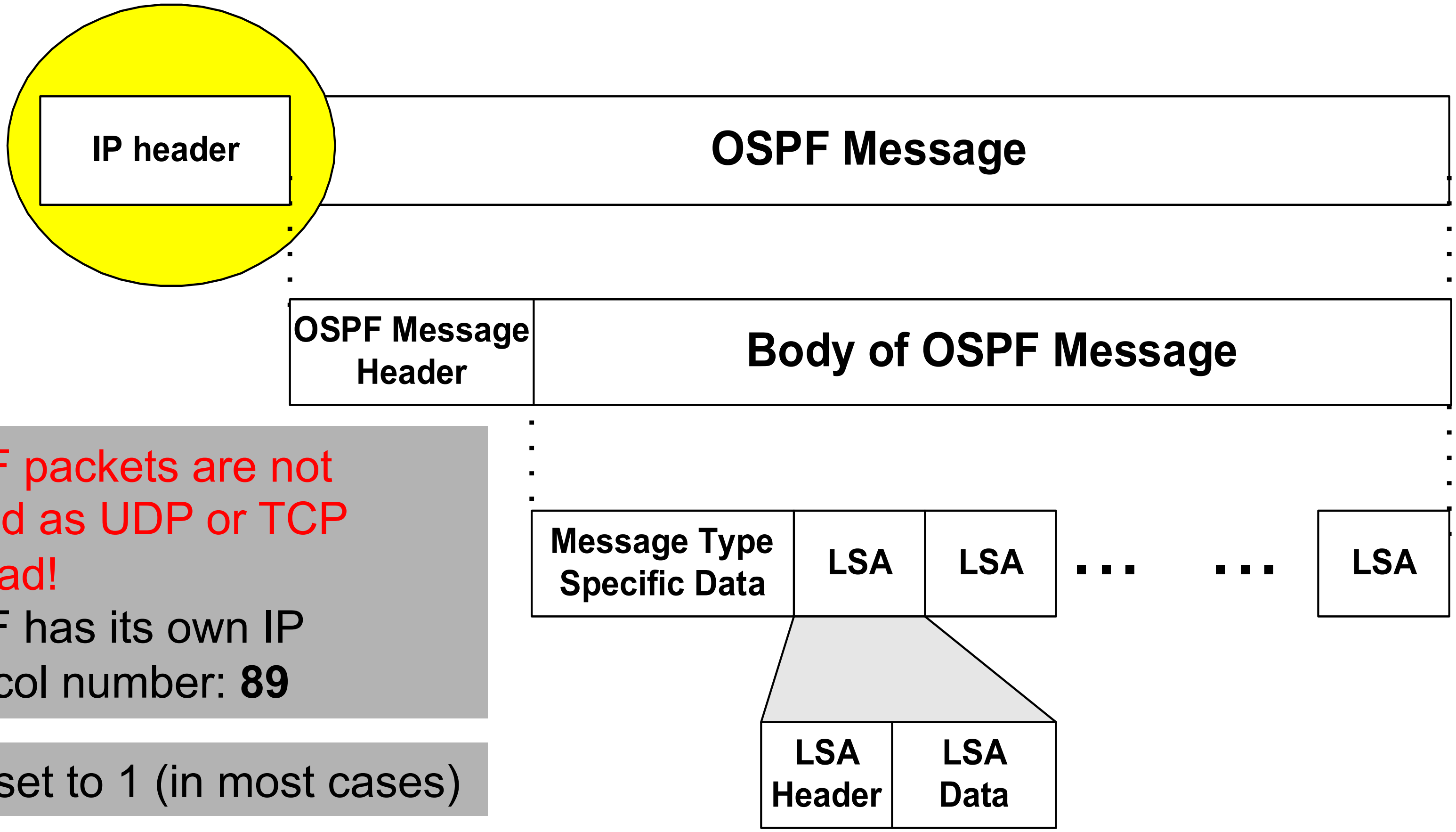
LSA
Entry in LSDB?
Yes
Sequence no different?
No
Ignore LSA
No
Yes
Step A
Add to LSDB
Sequence no higher?
Yes
Go to step A
No
Send LSAck
Send LSU with newer information to source
Flood LSA
End
Run SPF and calculate new routing table
End

OSPF Link State Packets

There are five types of Link-State Packets (LSPs).

1. hello: are used to establish and maintain adjacency with other OSPF routers. They are also used to elect the Designated Router (DR) and BackupDesignated Router (BDR) on multi-access networks.
2. database description (DBD or DD): contains an abbreviated list of the sending router's link-state database and is used by receiving routers to check against the local link-state database to make sure it has the latest information.
3. link-state request (LSR): used by routers to request more information about any entry in the DBD
4. link-state update (LSU): used to reply to LSRs as well as to announce new information. LSUs can contain 7 different types of Link-State Advertisements (LSAs)
5. link-state acknowledgement (LSAck): sent to confirm receipt of an LSU message

OSPF Packet Format



OSPF packets are not carried as UDP or TCP payload!
OSPF has its own IP protocol number: **89**

TTL: set to 1 (in most cases)

Destination IP: neighbor's IP address or 224.0.0.5 (ALLSPFRouters) or 224.0.0.6 (AllDRouters: (designated and backup designated only)

OSPF “advanced” features

- *security:* all OSPF messages authenticated (to prevent malicious intrusion)
- multiple same-cost paths allowed (only one path in RIP)
- for each link, multiple cost metrics for different TOS (e.g., satellite link cost set low for best effort ToS; high for real-time ToS)
- integrated uni- and multi-cast support:
  - Multicast OSPF (MOSPF) uses same topology data base as OSPF
- hierarchical OSPF in large domains.

Hierarchical OSPF